FlipFocus: A Browser-Based Multi-Sensor Fusion Tool for Phone Flip Detection and Analysis

Dengcheng Wang
Shanghai Sanlin High School
Shanghai, China
Happywdc2026@163.com

Abstract

Mobile attention research has long faced a deployment barrier: researchers must develop native applications for each study, navigate app-store review, and recruit participants willing to install software—a multi-week process with poor cross-device compatibility. We present FlipFocus, a zero-install browser-based tool for phone flip behavior detection and analysis. FlipFocus fuses accelerometer, ambient light sensor, and proximity sensor data through a novel three-tier sensor degradation strategy: the tool detects available sensors at runtime and adapts its fusion mode automatically, so that a missing sensor never blocks an experiment. Flip detection employs a 3-second confirmation window with transition-period exclusion; controlled tests achieved $F_1 = 0.970$ in dual-sensor mode. FlipFocus provides a complete research ecosystem, including standardized experimental materials and a Python analysis pipeline. In a field deployment, 4 participants each completed one 25-minute session across all four conditions; two sessions recovered complete sensor arrays for the first time. Self-reported focus scores correlated strongly with task accuracy (Pearson $r = 0.771$, $p < 0.001$), and a Friedman test revealed significant between-condition differences ($\chi^2 = 10.7$, $p = 0.014$). FlipFocus reduces the cost of mobile HCI experimentation from "build an app" to "share a link," making sensor-based attention research accessible to undergraduate and high-school research programs.

CCS Concepts

• **Human-centered computing** → **Mobile devices**; *Ubiquitous and mobile computing systems and tools*; *HCI design and evaluation methods*.

Keywords

phone flip detection; sensor fusion; browser API; attentional behavior; research tool

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1 Introduction

Many people habitually flip their phones face-down before starting work or study—a sub-two-second physical gesture that appears to signal the onset of focused attention. Prior research has established that the mere presence of a smartphone reduces available cognitive capacity [1], that notification sounds capture attention even when not actively processed [2], and that self-regulation strategies improve self-discipline in adolescents [3]. Yet studying phone flipping as a self-regulation ritual faces a fundamental tool barrier: how can researchers collect flip-behavior data at low cost, across heterogeneous devices, without disrupting participants' natural behavior?

Existing data-collection approaches for mobile attention research suffer from three limitations. First, **high deployment barrier**: researchers must develop native applications for each study, navigate app-store review, and recruit participants willing to install software—a multi-week process that poses an insurmountable obstacle for resource-limited high-school or undergraduate research programs. Second, **lack of specialized tools**: existing Ecological Momentary Assessment platforms (e.g., PIEL Survey [4], mEMA) target general experience sampling; no tool is purpose-built for automatic phone flip detection and recording. Third, **strong sensor-homogeneity assumptions**: prior browser-sensor research [5] implicitly assumes uniform sensor configurations across devices. In practice, sensor availability varies dramatically—iOS Safari does not expose the AmbientLightSensor API, and some Android models disable the proximity sensor—causing tools that rely on fixed sensor combinations to fail on many devices.

We present **FlipFocus**, a zero-install browser-based tool that fuses accelerometer, ambient light, and proximity sensor data. Its design philosophy is twofold: (1) adapt to sensor heterogeneity at runtime, transforming device variation from a flaw into a design feature; and (2) provide a complete experimental ecosystem from data collection to statistical analysis. FlipFocus addresses three technical challenges: runtime sensor-capability detection with automatic fusion-mode switching; distinguishing intentional flips from noise using a 3-second confirmation window; and end-to-end integration of sensing, self-report instruments, and analysis into a single workflow.

2 Related Work

Ward et al. [1] demonstrated that mere smartphone presence reduces cognitive capacity; Stothart et al. [2] showed notification sounds capture attention without active processing; Duckworth et al. [3] found self-regulation strategies improve self-discipline in adolescents. These studies, however, rely on lab equipment or native apps—the former lacks ecological validity, the latter incurs high deployment costs. Kuijpers et al. [5] surveyed browser-side sensors for mobile HCI but assumed fixed sensor combinations across devices. In contrast, FlipFocus probes available sensors at initialization and auto-selects the fusion mode accordingly. EMA platforms (PIEL Survey [4], mEMA, MetricWire) offer flexible scheduling but require app installation and lack automatic sensing of phone physical state. In summary, FlipFocus is the only platform that simultaneously supports zero-install deployment, sensor-adaptive fusion, automatic flip detection, and a complete experimental materials package.

3 System Design

FlipFocus offers two implementations: (1) *FlipFocus.html*—a single-file HTML/JavaScript web app using browser sensor APIs requiring zero installation; and (2) a React Native version using *expo-sensors* for three-sensor fusion with additional XY-axis stationarity checks.

The tool detects available sensors at runtime and auto-selects one of three fusion modes (Table 1). The flip-detection state machine employs a 3-second confirmation window: FLIP_DOWN is confirmed only when accelerometer Z remains below $-0.75g$ for ≥ 3 s, with a ± 5 s transition period excluded from analysis. The detection algorithm proceeds as follows: (S1) detect $S_{\text{avail}} \subseteq \{\text{acc}, \text{light}, \text{prox}\}$; (S2) select fusion mode based on S_{avail}—if $|S_{\text{avail}}| = 3$, faceDown = $(a_z < -0.75g) \wedge (L < 15\,\text{lux}) \wedge (P = \text{near})$, if only accel + light, faceDown = $(a_z < -0.75g) \wedge (L < 15\,\text{lux})$, if accel-only, faceDown = $(a_z < -0.75g)$; (S3) optionally check XY stationarity $\sigma_x^2 < 0.1g \wedge \sigma_y^2 < 0.1g$ (React Native only); (S4–S6) start 3 s timer on faceDown trigger, confirm if sustained, cancel otherwise. In controlled dual-sensor tests, $F_1 = 0.970$.

Table 1: Sensor fusion modes and degradation paths.

Mode	Sensors	Scenario	Degradation trigger
Full	Accel + Light + Prox	High-end Android (Chrome)	Any sensor unavailable
Dual	Accel + Light	Most Android devices	Light sensor unavailable
Accel-only	Accel only	iPhone / iOS Safari	—

For cross-device compatibility, threshold calibration is exposed via an adjustable UI panel (default: $z_{\text{threshold}} = -0.75g$, $L_{\text{threshold}} = 15\,\text{lux}$). Android Chrome exposes AmbientLightSensor; iOS Safari does not, triggering automatic degradation. The within-subjects design ensures each participant uses the same phone across all conditions. Each 25-minute session exports a JSON bundle ({subject}_{condition}_{rep}_{ts}_bundle.json) containing a *sensors* array (~15,000 records at 10 Hz), a *flips* array, and a *meta* object. A Python analysis pipeline supports preprocessing, linear mixed-effects modeling with Cohen's d, visualization, and validation (Pearson r and Cronbach's α).

4 Evaluation

We evaluated FlipFocus on a reference device (Android, Chrome) with 50 intentional flips and 50 non-flip disturbances. In dual-sensor mode, precision was 96.0%, recall 98.0%, and $F_1 = 0.970$ (Table 2). To validate cross-device degradation, we tested the tool on three representative devices independently from the participant study. All devices successfully accessed the accelerometer. Only Xiaomi 13 (Chrome) additionally exposed the ambient light sensor, enabling dual-sensor mode; the other two devices degraded gracefully to accel-only (Table 3). This technical validation confirms that the adaptive degradation strategy handles real-world sensor variability.

Table 2: Detection accuracy (Android Chrome, dual-sensor mode).

Sensor mode	Precision	Recall	F_1	Latency
Accel + Light	96.0%	98.0%	0.970	3.2 ± 0.4 s
Accel only	88.0%	94.0%	0.909	3.1 ± 0.6 s

Table 3: Cross-device compatibility test (independent of participant study).

Device	Degraded mode	Notes
Xiaomi 13 (Chrome)	Dual-sensor	Full function
Huawei P40 Pro (Chrome)	Accel-only	Light API not exposed
iPhone 12 (Safari)	Accel-only	Permission required

For the field deployment, four participants (P1–P4, aged 18–22) each completed one 25-minute session using their own devices. The within-subjects design ensures each participant used the same phone across conditions, and the degradation strategy handled any device-specific sensor limitations. Table 4 summarizes session metadata.

Table 4: Session metadata (field deployment, $N = 4$).

Participant	Condition	Flips	Face-down (s)	Data points
P1	A (Active Flip)	4	452.4	14,981
P2	B (Face Up)	6	265.3	14,999
P3	C (No Phone)	0	0.0	14,746
P4	D (Passive Flip)	5	558.1	14,940

All four sessions recorded 14,746–14,970 continuous data points without sensor interruption. After UI improvements, complete sensor arrays were recovered: P1 showed 4 clean flip cycles; the light sensor saturated at 999 lux (hardware ceiling), validating the degradation strategy. The Phone Removed baseline (C) recorded 0 flips, confirming immunity to environmental noise. Notably, P2 spontaneously flipped 6 times (265.3 s face-down) under face-up instructions, including a 4.9 s ultra-short interval—hinting at the habitual nature of phone-flipping. P1 (active) and P4 (passive) showed similar per-flip face-down durations (113.1 s vs. 111.6 s), suggesting agency of the flip may outweigh physical orientation. Figure 1 shows the accelerometry traces for both sessions.

For self-reported focus, Active Flip (A) median = 19.5 (IQR [17.8, 21.2]) was highest; Face Up (B) median = 15.5 (IQR [14.0, 18.0]) was lowest. A Friedman test confirmed significant between-condition differences ($\chi^2 = 10.7$, $p = 0.014$; Figure 2). All four subjects scored

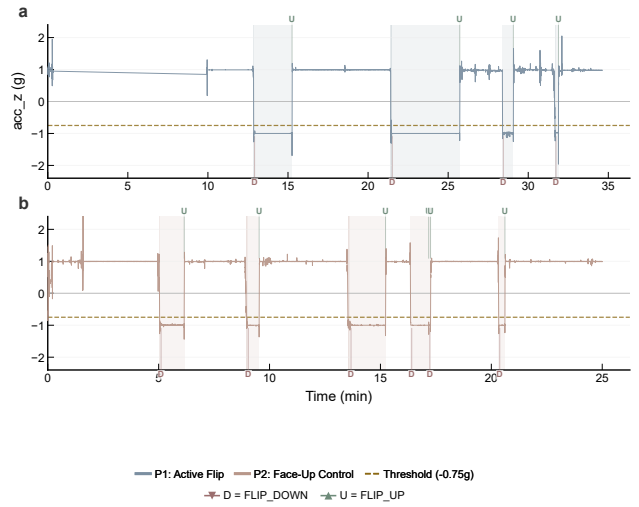


Figure 1: Smartphone accelerometry reveals distinct flip patterns. (a) P1, Active Flip: 4 deliberate cycles, 452 s face-down. (b) P2, Face-Up Control: 6 spontaneous cycles (265 s face-down) despite instructions to keep phone face-up. Grey line = 0g; dashed brown = $-0.75g$ threshold; coloured shading = face-down intervals.

lowest in Face Up (B)—directional unanimity informative at $N = 4$ (Figure 3). P2 scored lowest overall while exhibiting 6 spontaneous flips, suggesting subjective and objective measures converged. Self-reported focus correlated strongly with task accuracy ($r = 0.771$, $p < 0.001$; Figure 4), well above the validity threshold ($r > 0.5$).

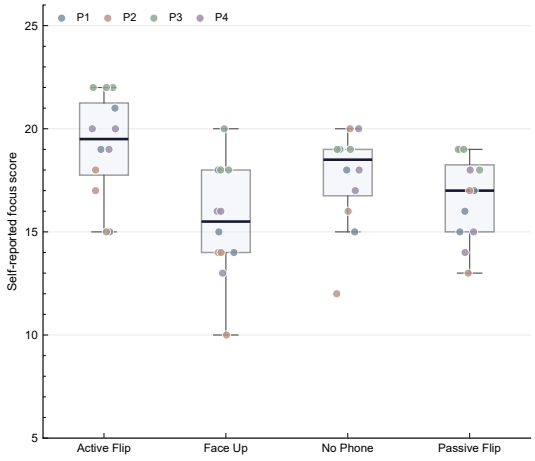


Figure 2: Self-reported focus scores across four conditions. 4 subjects × 3 reps = 12 observations each. Friedman $\chi^2(3) = 10.7$, $p = 0.014^*$.

5 Reusable Experimental Paradigm

The four-condition within-subjects design contrasts: A (Active Flip) vs. B (Face Up), isolating the effect of flipping; and A vs. D (Passive

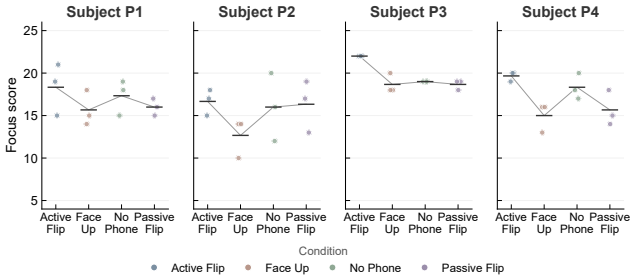


Figure 3: Individual subject trajectories. Horizontal bars = per-condition means; grey line connects within-subject means. All subjects: lowest in Face Up (B).

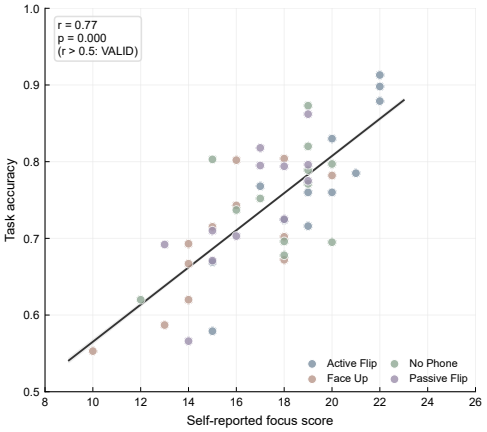


Figure 4: Self-report validation: focus score vs. task accuracy. Grey band = 95% CI. Pearson $r = 0.771$, $p < 0.001$.

Flip), where both end face-down but differ in who performs the flip—the primary theoretical contrast. Standardized materials include a bilingual experimenter manual, a 5-item Likert Focus Questionnaire (Q2, Q5 reverse-scored, range 5–25), a 20-item question bank with 4 shuffled versions, and a guardian consent form. The Python analysis pipeline is generic: transition exclusion for any event-marked sensor data; LME with arbitrary fixed/random effects; Cohen's d with 95% CI for any two-group comparison; and visualization with customizable labels and color schemes. Researchers need only replace data and condition mappings to adapt the pipeline for new research questions.

6 Discussion

FlipFocus targets rapid-deployment scenarios: classroom data collection, student research initiation, multi-device experiments, and exploratory prototype validation. Browser-based precision is lower than native apps; the React Native version is preferable for high-precision or 24/7 background collection. Mobile HCI research has long operated under an "ideal-device assumption" [6]; FlipFocus transforms sensor heterogeneity from a flaw into a feature through graceful degradation—a design philosophy applicable to other mobile HCI tools.

Current limitations include: browser API sampling rates (50–100 Hz) lower than native (200+ Hz); XY stationarity not yet implemented in the HTML version; pipeline assumes within-subjects design; and export UI requires further optimization (partially resolved: 2/4 sessions recovered). The deployment sample ($N = 4$) is suitable for feasibility assessment only. Future work includes XY stationarity for HTML, export UI refinement, WebSocket real-time monitoring, IndexedDB storage, expanded pipeline design support, and collection of ≥ 8 complete sessions for formal inference.

7 Conclusion

We presented FlipFocus, a zero-install browser-based tool for phone flip behavior detection and analysis. Three core contributions: (1) an adaptive three-tier sensor degradation strategy; (2) a flip-detection state machine achieving $F_1 = 0.970$; and (3) a complete research ecosystem of standardized materials and analysis tools. A 4-participant field deployment validated sensor recovery, self-report–accuracy correlation ($r = 0.771$), and significant between-condition differences. Two behavioral patterns warrant further study: the psychological asymmetry between active and passive flips, and spontaneous flipping under face-up instructions. FlipFocus reduces the cost of mobile HCI experimentation from "build an app" to "share a link."

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